In [1]:
```
# importing major libraries
import numpy as np
import pandas as pd
import plotly.express as px

# addition library
import warnings
warnings.filterwarnings('ignore')
```

In [2]:
```
df = pd.read_csv('CO2.csv').iloc[:,2:]
```

In [3]:
```
df.head()
```

Out[3]:

	Volume	Weight	CO2
0	1000	790	99
1	1200	1160	95
2	1000	929	95
3	900	865	90
4	1500	1140	105

In [4]:
```
px.scatter_3d(df,x='Volume',y='Weight',z='CO2',color='CO2').show()
```

In [5]:
```
df
# x-y
# x - Independent columns
# y - dependent columns
x = df.iloc[:,0:2]
y = df.CO2  #2d array # Data frame
```

In [6]:
```
# Train_Test_Split
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [7]:
```
# Linear Regression
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr
```

Out[7]:
```
▾ LinearRegression
LinearRegression()
```

In [8]:
```
print('rows for training',x_train.shape[0])
print('rows for testing',x_test.shape[0])
```
```
rows for training 28
rows for testing 8
```

In [9]:
```
lr.fit(x_train,y_train)
```
Out[9]:
```
▾ LinearRegression
LinearRegression()
```

In [10]:
```
y_pred=lr.predict(x_test)
```

In [11]:
```
y_test-y_pred
```
Out[11]:
```
35    13.837699
13    -7.152581
26    -0.783275
30     8.862316
16    -3.062149
31    10.156109
21    -5.284220
12     1.546767
Name: CO2, dtype: float64
```

In [12]:
```
from sklearn.metrics import r2_score

r2_score(y_test,y_pred)
```
Out[12]:
```
0.32941109624012743
```

In [13]:
```
# Underfitting
lr.score(x_train,y_train)
```
Out[13]:
```
0.2595386654370385
```

In [14]:
```
lr.coef_
```
Out[14]:
```
array([0.00428741, 0.00804928])
```

In [15]:
```
lr.intercept_
```
Out[15]:
```
np.float64(84.21502006695282)
```

In [16]:
```
m1 = 0.00428741
m2 = 0.00804928
int = 84.21502006695282
```

In [17]:
```
x1=2500
x2=1395
```

```
m1*x1+m2*x2+int
```

Out[17]: 106.16229066695283

In [18]:
```
y_pred[0]
```

Out[18]: np.float64(106.16230071166162)

In [19]:
```
x_test.iloc[0]
```

Out[19]:
```
Volume    2500
Weight    1395
Name: 35, dtype: int64
```

In [20]:
```
from sklearn.metrics import mean_absolute_error,mean_squared_error,root_mean_s
mean_absolute_error(y_test,y_pred)
mean_squared_error(y_test,y_pred)
root_mean_squared_error(y_test,y_pred)
```

Out[20]: 7.620976698335832

In [21]:
```
from sklearn.metrics import mean_absolute_error,mean_squared_error,root_mean_s
mean_absolute_error(y_test,y_pred)
mean_squared_error(y_test,y_pred)
root_mean_squared_error(y_test,y_pred)
r2_score(y_test,y_pred)
```

Out[21]: 0.32941109624012743

In [22]:
```
from sklearn.metrics import r2_score
r2=r2_score(y_test,y_pred)
n=x_train.shape[0]  #use training data
k=x_train.shape[1]
adjusted_r2=1-(1-r2)*((n-1)/(n-k-1))
adjusted_r2
```

Out[22]: 0.27576398393933754

In [23]:
```
import seaborn as sns
mpg = sns.load_dataset('mpg')
mpg.head()
#Q ---> Predict Mileage
```

Out[23]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year
0	18.0	8	307.0	130.0	3504	12.0	70
1	15.0	8	350.0	165.0	3693	11.5	70
2	18.0	8	318.0	150.0	3436	11.0	70
3	16.0	8	304.0	150.0	3433	12.0	70
4	17.0	8	302.0	140.0	3449	10.5	70

In [24]:
```
mpg.isnull().mean()
```

Out[24]:
```
mpg             0.000000
cylinders       0.000000
displacement    0.000000
horsepower      0.015075
weight          0.000000
acceleration    0.000000
model_year      0.000000
origin          0.000000
name            0.000000
dtype: float64
```

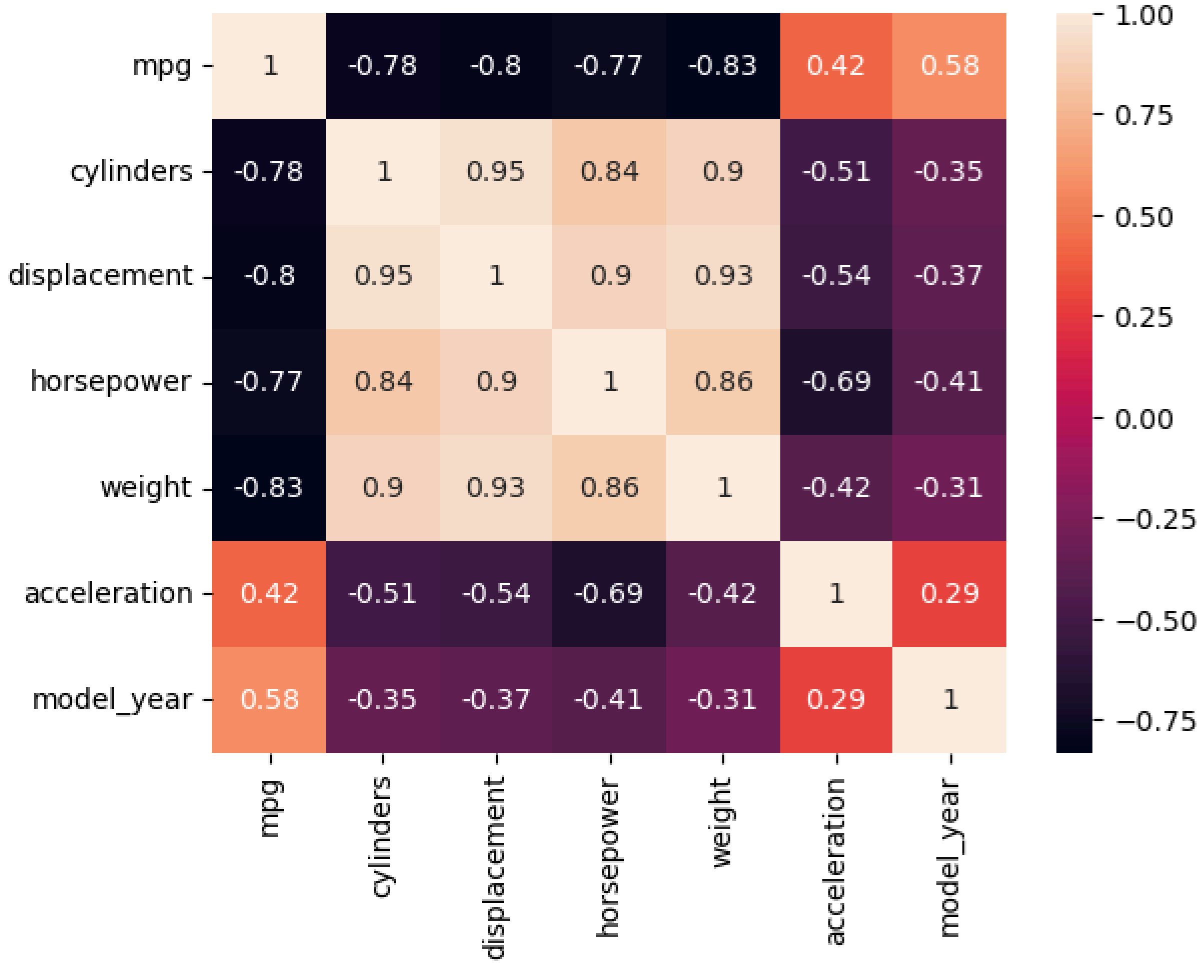
In [25]:
```
mpg['horsepower'].fillna(mpg['horsepower'].median(), inplace=True)
```

In [26]:
```
mpg.isnull().sum()
```

Out[26]:
```
mpg             0
cylinders       0
displacement    0
horsepower      0
weight          0
acceleration    0
model_year      0
origin          0
name            0
dtype: int64
```

In [27]:
```
sns.heatmap(mpg.corr(numeric_only=True),annot=True)
```

Out[27]:
```
<Axes: >
```



In [28]:

```
# x-y
# x - Independent columns
# y - dependent columns
x = mpg.iloc[:,1:7]
y = mpg.mpg  #2d array # Data frame
```

In [29]:

```
# Train_Test_Split
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test = train_test_split(x,y,test_size=0.2,random_stat
```

In [30]:

```
# Linear Regression
from sklearn.linear_model import LinearRegression
lr = LinearRegression()
lr
```

Out[30]:

```
▾ LinearRegression
LinearRegression()
```

In [31]:

```
print('rows for training',x_train.shape[0])
print('rows for testing',x_test.shape[0])
```

```
rows for training 318
rows for testing 80
```

In [32]:
```
lr.fit(x_train,y_train)
```

Out[32]:

```
▾ LinearRegression
LinearRegression()
```

In [33]:
```
y_pred=lr.predict(x_test)
```

In [34]:
```
y_test-y_pred
```

Out[34]:
```
198    1.469575
396   -2.710936
33    -2.759651
208   -4.004440
93     1.386835
         ...
249   -2.733775
225   -3.159055
367   -2.945680
175   -0.502433
285   -3.211445
Name: mpg, Length: 80, dtype: float64
```

In [35]:
```
from sklearn.metrics import r2_score

r2_score(y_test,y_pred)
```

Out[35]:
```
0.8244245330943358
```

In [ ]:
```
```